Baseline Project Plan: Green Valley Hospital Management Information System (GVH-MIS)

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Course: System Analysis and Design

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Project Repository: https://github.com/J-ace1/Green-Valley-Hospital-Information-Management-System.git

1. Introduction

i. Brief Overview

Green Valley Hospital is a medium-size hospital that provides both outpatient and inpatient services, lab facilities, pharmacies, and emergency services. Green Valley Hospital uses paper and manual systems. This project aims to introduce a comprehensive HMIS to Green Valley Hospital within 7 months at an estimated cost of KES 3,500,000. This plan forms the baseline for measuring the progress made throughout the project.

ii. Problem and Opportunity

Problem:

- Lack of a centralized patient's database

- Little interdepartmental data exchange

Consequences:

- Duplicates/loss of patient information affecting continuity of treatment;

- Long time spent on manual registration and searching for files;

- Delays in obtaining patients' history and lab tests results;

- Billing mistakes and loss of money;

- Bad stock control, resulting in shortage of medicines and their waste;

- Late and erroneous management reports.

Opportunity: An integrated system will help the hospital provide faster service to patients and recover their lost money through utilization of data.

iii. System Type

The system is a **Management Information System (MIS)** built on a **Transaction Processing System (TPS)** foundation. It captures daily transactions (registrations, orders, dispensing, payments) and turns them into summary reports and dashboards for management.

iv. Development Approach

The project adopts the Systems Development Life Cycle (SDLC) tailored to suit packaged software and involves prototyping during configuration. The life cycle stages include planning, analysis, design, implementation, and maintenance. The departments evaluate the prototypes of their screens and workflows before actual configuration, thus minimizing the chances of implementing systems that users reject.

v. Recommended Course of Action

Acquire and configure a commercially available HMIS solution that is known to work well, rather than developing a solution from scratch or using manual systems.

Lowest risk – it has been tried before in other hospitals; there's a danger of problems if it is developed internally.

Fastest to deliver – configuring a product to fit within the 7-month deadline.

Affordable – KES 3,500,000 is feasible; internal development would cost more.

Vendor support – training and maintenance are included. Next step – management signs off on this recommendation to move into requirements and vendor selection.

vi. Project Scope

Units affected: Registration/Records, Outpatient, Inpatient, Laboratory, Pharmacy and Stores, Billing/Finance, and Management.

Interaction with other systems:

- Laboratory analyzers (results feed into patient records)
- The hospital's accounting package (billing data export)
- Mobile money payments such as M-Pesa (payment recording)
- Insurer and government claims (e.g., SHA) are out of scope for integration. The system produces claim reports for manual submission.

- **Range of system capabilities:** patient registration and electronic records, appointments, outpatient and inpatient management, laboratory, pharmacy and inventory, billing and finance, and management reports and dashboards.

Objectives:

- Implement a centralized electronic patient record by end of Month 6 (go-live).
- Reduce registration and waiting time by 40% within 1 month of go-live.
- Reduce billing errors by 60% through automated charge capture.
- Provide real-time drug and supply tracking and on-demand management reports.
- Complete the project within 7 months and KES 3,500,000.

Deliverables: requirements document, system design, configured and tested HMIS, migrated data, trained staff, user manuals, and a closure report.

Out of scope: hardware beyond what the system needs, telemedicine and mobile apps, external insurer and government integration, digitization of records older than 5 years, and changes to clinical procedures or staffing structure.

2. System Description

i. Alternative Solutions

Option	Description	Assessment
A. Improve manual process	Better filing and standalone spreadsheets	Cheap, but does not solve the core problems
B. Build a custom system	In-house or contracted development	Fits the needs exactly, but is slow, costly, and risky for 7 months
C. Commercial HMIS (recommended)	Buy and configure a proven system	Fastest, reliable, affordable, and supported by the vendor

ii. Narrative Description

HMIS is a role-based system that can be accessed through workstations available within every department. Patients are entered into the system only once; each time a patient is seen again, all his visits, diagnoses, lab results, prescriptions, and bills will be recorded in the same file using one ID number for each patient. Doctors get to see the complete history of the patient at once; lab and pharmacy orders are transmitted electronically, and billing is automated based on services provided.

3. Feasibility Assessment

i. Project Costs and Benefits

Item	Cost (KES)
Software licensing and customization	1,400,000
Hardware (servers, workstations, network)	800,000
Project team, management, and consulting	450,000
Staff training	300,000
Data migration	250,000
Contingency (about 9%)	300,000
Total	3,500,000

Benefits: recovered revenue, lower paper costs, less manual work, better patient safety, faster service, and better decisions. Assuming about KES 1,500,000 per year in recovered revenue and savings, payback is roughly 2.3 years.

ii. Feasibility by Type

Type	Assessment
Economic	Cost is justified by recovered revenue and savings, with payback in about 2.3 years.
Operational	Staff and management need and support the system. Resistance is managed through early involvement and training.
Technical	Commercial HMIS products are proven. Difficulties: data migration, unreliable power or internet (backup power, local server), and old lab equipment.
Schedule	7 months is achievable with a packaged system.
Legal and contractual	Must comply with the Data Protection Act, 2019. The vendor contract fixes price, deliverables, support, and data ownership.
Political	Management sponsors the project. Departments are involved early to gain support.

iii. High-Level Schedule

Month	Activity
1	Planning and requirements analysis
2	System design and vendor selection
3-4	Configuration, customization, and hardware installation

5	Data migration and testing (including user acceptance)
5-6	Staff training; go-live at end of Month 6
7	Post-implementation support and project closure

4. Management Issues

i. Management Concerns

Concern	Response
Disruption to patient services	Phased go-live, parallel run with manual system for a short period
Staff resistance	Early involvement, training, and department champions
Cost overruns	Fixed-price vendor contract, 9% contingency, monthly budget reviews
Data security and privacy	Role-based access, audit logs, regular backups
Dependence on the vendor	Support agreement and handover documentation

ii. Team Composition

Role	Responsibility
Project Sponsor (Hospital Management)	Approves scope, budget, and major decisions
Project Manager	Plans, tracks, and reports on the project
Systems Analyst	Gathers requirements and designs workflows
Vendor Technical Team	Installs, configures, and supports the system
Department Representatives	Provide requirements, test, and champion the change
Trainer	Delivers staff training

iii. Communication Plan

Progress reports go to the Sponsor **monthly;** the project team meets **weekly**, department demos and updates run **bi-weekly**, and issues are escalated to the Sponsor **as needed**.

iv. Project Standards and Procedures

Scope, cost, and schedule changes need written Sponsor approval. Requirements, designs, and test results use standard templates with version control. Each module must pass user acceptance testing before go-live. Progress is measured against this baseline, and all patient data handling follows the Data Protection Act, 2019.